MEMORANDUM

Travel Behavior and Forecasting
Harvard University Graduate School of Design

To:	Carole Voulgaris
From:	Hannah Jun
Date:	March 3, 2025
Re:	Recommended format and style for professional memos

Note: Relevant datasets and code that were use used to produce the analysis summarized in this memo can be found at https://github.com/urbanbyline/ses5394_t1_team5

The purpose of this memo is to demonstrate and describe a recommended format and style for a professional memo. Beginning a memo with a clear, concise statement of purpose is a best practice.

Key findings:

- Chauffeured trips makes a small percentage of daily trips made
- Specific factors like gender and age influence if you make a chauffeured trip and how many
- Household factors like size and vehicle access also influence if a chauffeured trip will be made.

Background

Chauffeured trips are trips in which someone is picking or dropping off someone else. They are interesting trips to analyze as they have unique characteristics to transportation:

- They inherently have more than 1 person in the vehicle vs the overarching pattern of single occupancy vehicles
- They don't neatly fit into discretionary or non-discretionary trip given the trip purpose changes by user
- Why you would pick up or drop off someone is a relational decision

Data

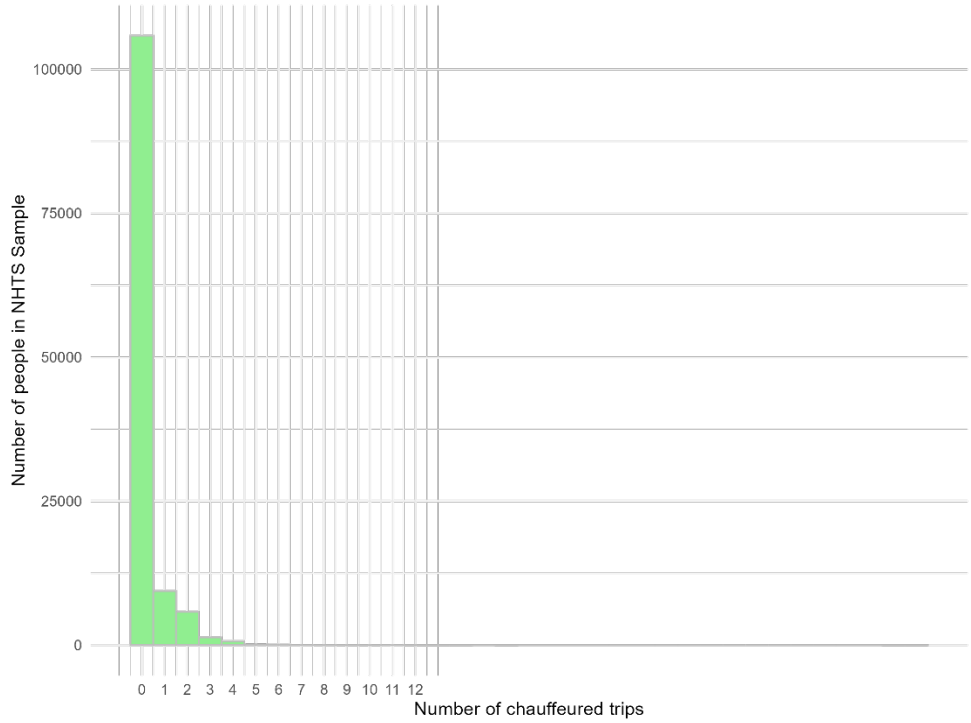
Using the NHTS 2017 datasets on trip, household level, and person level, I constructed several coefficient variables that I hypothesized could have an impact on the number of chauffeured trips taken.
On a household level I considered the following:

1. Household size – HHSIZE
2. Vehicle availability – veh_avail
3. Household income – income
4. Number of Seniors in the household – n_seniors_in_hh
5. Number of children in the household – n_child
6. Number of workers in the household - WRKCOUNT

On a person level I considered the following:

1. Age – R_AGE
2. Gender - female
3. Part-Time Work – part_time
4. Day of Travel – TRAVDAY

I defined chauffeured trips as trips which were labeled with category 06 "Drop off /pick up someone" for WHYTO but this might be a limited approach given trips can be multipurpose. For example visiting friends or relatives to then drop off someone to their house cannot be accurately captured.



Graph 1: Count of chauffeured trips taken daily by NHTS sample

Graph 1 visualizes the number of chauffeured trips within the entire dataset and shows on any given day most people are making zero chauffeured trips. Given the high zero count and that the average count of chauffeured trips sits at 0.244 with a higher standard deviation of 0.728 a zero inflated regression model would let us see what factors impact people to make a chauffeured trip at all and then what influences people to make more chauffeured trips.

Results

The zero-inflated choice regression model shows us certain coefficients strongly influence chauffeured trips. In Table 1, we see that the p-values that are highlighted in grey as statistically significant and the coefficients highlighted in yellow as having a strong positive or negative value to impact the model. For example, we see the number of children not only decrease the probability of taking zero trips but also the number of chauffeured trips. Considering that the majority of children do not have a license to drive or an available car, it makes sense that their presence increases chauffeured trips. Vehicle availability also impacts the ability to make chauffeured trips with having no vehicles increasing zero trips and reducing the number of chauffeured trips. We see that gender has a split impact with being female increasing the number of chauffeured trips but they are also less likely to take zero trips likely meaning that women are overrepresented at the extremes. Seeing the interaction impact between n_child, women, and marriage status could help parse out this detail.

Table 1. Results of a zero- inflated choice regression model predicting chauffeured

	Count Model Interpretation		Zero-Inflation Model Interpretation	
	Coefficient estimate	p-value	Coefficient estimate	p-value
Intercept	-1.033680	2e-16	0.156398	0.269
n_seniors_in_hh	-0.011355	0.693242	0.037680	0.373
n_child	0.477606	2e-16	-3.842762	2e-16